

Econometrics II – Assignment 4:

Volatility Timing in the U.S. Stock Market

Spring 2026

1 Introduction

Using monthly U.S. log excess returns $x_t = \text{mkt}_t - \text{rf}_t$ from July 1926 to February 2026, this report asks (i) whether x_t exhibits ARCH effects and asymmetric (leverage) volatility, and (ii) whether scaling returns by a conditional standard deviation from a fitted volatility model outperforms the expanding-window benchmark in equation (2.1) of the assignment. Following Harvey et al. (2018) and Moreira and Muir (2017), volatility targeting improves Sharpe ratios when conditional means do not move proportionally to conditional volatility. A GJR-GARCH(1,1) with Student- t innovations, estimated by (quasi-)maximum likelihood, is preferred. ARCH effects are massively significant, the leverage parameter is significant, and the resulting scaled return x_t^* delivers an annualized Sharpe ratio of 0.43 versus 0.34 for the unscaled market and 0.35 for the expanding-window benchmark; an HAC spanning regression rejects $\alpha = 0$ for x_t^* but not for the benchmark.

2 Description of the Data

The data are monthly, in percent per month, from Kenneth French’s Data Library: mkt_t is the log return on the value-weighted CRSP market portfolio and rf_t is the log one-month T-bill rate, both transformed from the published simple returns as continuously compounded equivalents. The sample contains $T = 1,196$ observations, 1926M7–2026M2. We work throughout with the log excess return $x_t = \text{mkt}_t - \text{rf}_t$.

The series x_t has mean 0.55%, standard deviation 5.30%, skewness -0.52 and excess kurtosis 6.48 already signalling fat tails. Figure 1 shows that x_t has near-zero serial correlation, while $|x_t|$ and x_t^2 display strong, slowly decaying autocorrelation: the classic volatility-clustering pattern. Large clusters coincide with 1929–32, 1987, 2008 and 2020.

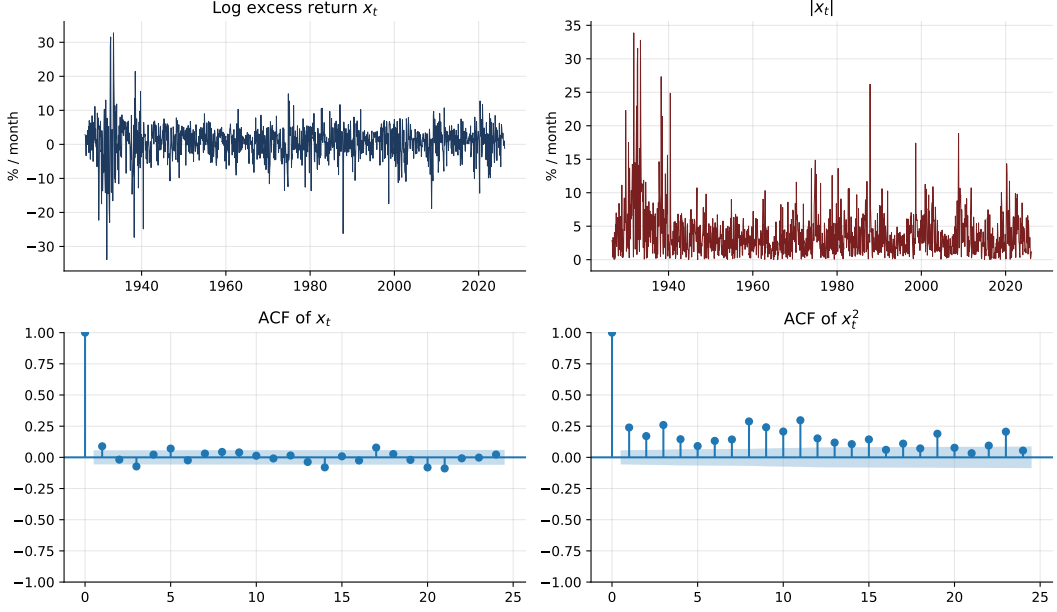


Figure 1: Top: x_t and $|x_t|$. Bottom: sample ACFs of x_t and x_t^2 . The flat ACF of x_t and the persistent ACF of x_t^2 motivate a constant-mean ARCH-type model.

3 Econometric Theory

Model. Given the absence of autocorrelation in x_t , we use a constant-mean specification

$$x_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{iid}(0, 1), \quad (1)$$

where $\sigma_t^2 = \mathbb{V}[x_t \mid \mathcal{I}_{t-1}]$ depends on past information. We compare nested specifications for σ_t^2 :

$$\text{ARCH}(q) : \quad \sigma_t^2 = \omega + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}^2, \quad (2)$$

$$\text{GARCH}(1, 1) : \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (3)$$

$$\text{GJR-GARCH}(1, 1) : \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \kappa \varepsilon_{t-1}^2 \mathbb{I}(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2. \quad (4)$$

For positive variances we require $\omega > 0$ and the slope parameters non-negative. Covariance stationarity requires $\sum_j \alpha_j < 1$ in (2), $\alpha + \beta < 1$ in (3) and, with $\mathbb{E}[\mathbb{I}(\varepsilon_{t-1} < 0)] = 1/2$ under a symmetric distribution, $\alpha + \beta + \frac{1}{2}\kappa < 1$ in (4). The unconditional variance for GARCH(1,1) is $\sigma^2 = \omega / (1 - \alpha - \beta)$. In (4) a positive κ encodes the *leverage effect*: negative shocks raise next-period variance by $(\alpha + \kappa)\varepsilon_{t-1}^2$ versus $\alpha\varepsilon_{t-1}^2$ for positive shocks.

Estimator. Stacking the parameter vector θ , we estimate by (quasi-)maximum likelihood, maximising the Gaussian log-likelihood

$$\ell_T(\theta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log \sigma_t^2(\theta) + \frac{(x_t - \mu)^2}{\sigma_t^2(\theta)} \right]. \quad (5)$$

When z_t is non-Gaussian the Gaussian likelihood is mis-specified but the MLE remains a QMLE: under regularity (correct conditional-variance specification, stationarity, finite fourth moments of ε_t , and an identification condition on θ_0) it is consistent and asymptotically normal,

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}^{-1} \mathcal{J} \mathcal{I}^{-1}),$$

with $\mathcal{I}$ the expected Hessian and $\mathcal{J}$ the expected outer product of scores; under Gaussian z_t , $\mathcal{J} = \mathcal{I}$ and the variance simplifies to $\mathcal{I}^{-1}$. We report the sandwich SEs. Following the course we also estimate the model with $z_t \sim t(\nu)/\sqrt{\nu/(\nu-2)}$ (scaled Student- t); this delivers fully efficient ML inference when innovations are leptokurtic.

Hypotheses and tests. For the existence of ARCH effects we use the Engle/Breusch–Pagan LM test on $\hat{\varepsilon}_t$ from a mean-only regression: under H_0 of conditional homoskedasticity in (2), $\zeta_{\text{ARCH}} = TR^2 \xrightarrow{d} \chi^2(h)$ for the auxiliary regression of $\hat{\varepsilon}_t^2$ on h of its own lags. Asymmetry is tested with $H_0 : \kappa = 0$ in (4) via a robust t -test, and by likelihood ratio against GARCH(1,1). Standardized residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$ from the fitted model are inspected with (i) Ljung–Box on $\hat{z}_t$ and $\hat{z}_t^2$ (null: no autocorrelation; $\xi_{LB} \sim \chi^2(h)$), (ii) a repeat ARCH-LM test on $\hat{z}_t$ and (iii) the Jarque–Bera normality test. Finally, the spanning regression

$$x_t^* = \alpha + \beta x_t + u_t \quad (6)$$

is estimated by OLS with Newey–West HAC SEs at $\lfloor 4(T/100)^{2/9} \rfloor = 6$ lags; a positive, significant $\hat{\alpha}$ indicates that volatility timing improves the unconditional Sharpe ratio.

4 Empirical Analysis

Preliminary tests. The ARCH-LM test on mean-only residuals rejects H_0 overwhelmingly at every horizon: $\zeta_{\text{ARCH}} = 65.3, 134.4, 236.7$ at $h = 1, 5, 12$, all with $p < 10^{-15}$. ARCH effects are unambiguous.

Model selection (general-to-specific). Table 1 reports five nested candidates. ARCH(1) fails the LB($\hat{z}^2$) and residual ARCH tests. ARCH(5) absorbs most clustering but is over-parameterised. GARCH(1,1) is a strict improvement on AIC (7063.4 vs. 7134.7) and BIC, and its standardized residuals comfortably accept the null of no remaining ARCH structure (LB($\hat{z}^2$, 12) $p = 0.97$, ARCH(12) $p = 0.97$); persistence $\hat{\alpha} + \hat{\beta} = 0.98$ is high

but < 1 . Adding asymmetry yields GJR-GARCH(1,1) with $\hat{\kappa} = 0.125$ ($t = 1.79$) and an LR statistic of $12.0 \sim \chi^2(1)$, $p < 0.001$. Replacing Gaussian innovations by Student- $t(\nu)$ gives a large fit gain ($\Delta \log L = 34.3$, $\Delta \text{AIC} = -66.6$), $\hat{\nu} \approx 7.1$ (decisively non-Gaussian) and a sharper asymmetry estimate $\hat{\kappa} = 0.179$ ($t = 2.93$, $p = 0.003$). We select GJR-GARCH(1,1)- t as the preferred model.

Table 1: Model comparison. Robust (sandwich) SEs in parentheses; (Q)MLE, constant mean. Persistence is $\sum_j \alpha_j$ for ARCH(q), $\alpha + \beta$ for GARCH(1,1), $\alpha + \beta + \frac{1}{2}\kappa$ for GJR. Diagnostics: Ljung–Box at $h=12$ on $\hat{z}_t$ and $\hat{z}_t^2$, ARCH-LM at $h=12$ on $\hat{z}_t$, and Jarque–Bera on $\hat{z}_t$.

Parameter	ARCH(1)	ARCH(5)	GARCH(1,1)	GJR(1,1)	GJR(1,1)-t
μ	0.732 (0.157)	0.694 (0.145)	0.716 (0.125)	0.639 (0.120)	0.858 (0.123)
ω	21.966 (2.610)	10.498 (2.521)	0.734 (0.292)	1.029 (0.516)	1.567 (0.554)
α	0.205	0.653	0.135 (0.026)	0.055 (0.035)	0.030 (0.022)
κ (asym.)	–	–	–	0.125 (0.070)	0.179 (0.061)
β	–	–	0.845 (0.028)	0.840 (0.039)	0.803 (0.046)
ν	–	–	–	–	7.10 (1.548)
Persistence	0.205	0.653	0.980	0.957	0.922
$\log L$	-3646.8	-3560.4	-3527.7	-3521.7	-3487.4
AIC	7299.6	7134.7	7063.4	7053.5	6986.9
BIC	7314.8	7170.3	7083.7	7078.9	7017.4
LB($\hat{z}$, 12) p	0.085	0.345	0.299	0.252	0.320
LB($\hat{z}^2$, 12) p	<0.001	0.490	0.968	0.946	0.949
ARCH(12) p	<0.001	0.459	0.969	0.944	0.944
JB p	<0.001	<0.001	<0.001	<0.001	<0.001

Misspecification of the preferred model. For GJR- t the standardized residuals show no autocorrelation ($\text{LB}_{12} p = 0.32$), no residual ARCH ($p = 0.95$ and 0.94), and the implied persistence $\hat{\alpha} + \hat{\beta} + \frac{1}{2}\hat{\kappa} = 0.922$ satisfies covariance-stationarity. Jarque–Bera rejects normality of $\hat{z}_t$ ($p < 0.001$) exactly the leptokurtosis the t distribution accounts for, so QMLE-with-sandwich inference and the parametric t likelihood both remain valid. Economically, the preferred estimates ($\hat{\mu} = 0.86\%$ /month, $\hat{\beta} = 0.80$, $\hat{\alpha} = 0.03$, $\hat{\kappa} = 0.18$) imply that a -5% shock raises next-period σ_t^2 roughly $(\hat{\alpha} + \hat{\kappa})/\hat{\alpha} \approx 7$ times more than a $+5\%$ shock the leverage/volatility-feedback pattern.

Volatility-targeting performance. Using $\hat{\sigma}_t$ from the preferred model we form $x_t^* = (\bar{\sigma}/\hat{\sigma}_t)x_t$ with $\bar{\sigma} = 10/\sqrt{12} \approx 2.89$ (percent). Figure 2 contrasts $\hat{\sigma}_t$ with the expanding-

window benchmark $\hat{\sigma}_t^{ew}$: the GJR- t estimate reacts much more sharply to clustering episodes. Over the common sample (1928M7–2026M2, $T = 1,172$), Table 2 shows the annualized Sharpe ratio rising from 0.34 for x_t to 0.43 for x_t^* , while the benchmark gives only 0.35; the worst-month drawdown improves to -19.6% for x_t^* vs. -33.9% for x_t . Figure 3 confirms a higher and smoother cumulative return for x_t^* . The spanning regression (6) in Table 3 formalises the comparison: $\hat{\alpha} = 0.095$ ($t = 2.50$, $p = 0.013$) for x_t^* , so we reject $H_0 : \alpha = 0$ GJR- t scaling adds genuine risk-adjusted value beyond the unscaled market. For the expanding-window benchmark $\hat{\alpha} = 0.011$ ($t = 0.59$, $p = 0.56$): the null is not rejected, consistent with the benchmark being too slow-moving relative to actual conditional volatility.

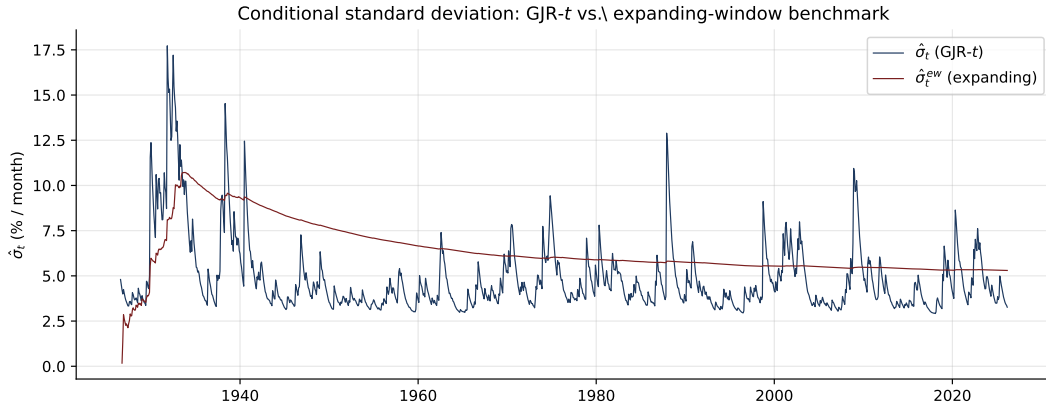


Figure 2: Conditional standard deviation: GJR- t vs. expanding-window benchmark.

Table 2: Performance over the common sample (1928M7–2026M2, $T=1,172$). Sharpe annualized; returns in % per month.

Series	Mean	Std	Sharpe (ann.)	Skew	Ex. Kurt	Min	Max
x_t (unscaled)	0.525	5.327	0.342	-0.52	6.45	-33.87	32.77
x_t^* (GJR- t)	0.360	2.915	0.428	-0.97	3.36	-19.56	8.87
xs_scaled_bt	0.238	2.376	0.347	-0.90	5.08	-15.74	9.74

Table 3: Spanning regression $x_t^* = \alpha + \beta x_t + \varepsilon_t$, OLS with Newey–West HAC SEs (6 lags). $H_0 : \alpha = 0$.

	x_t^* (GJR- t)	xs_scaled_bt
α	0.0950	0.0106
HAC SE	(0.0380)	(0.0181)
t	2.50	0.59
p	0.013	0.558
β	0.5043	0.4329
HAC SE	(0.0336)	(0.0165)
R^2	0.849	0.942
T	1172	1172

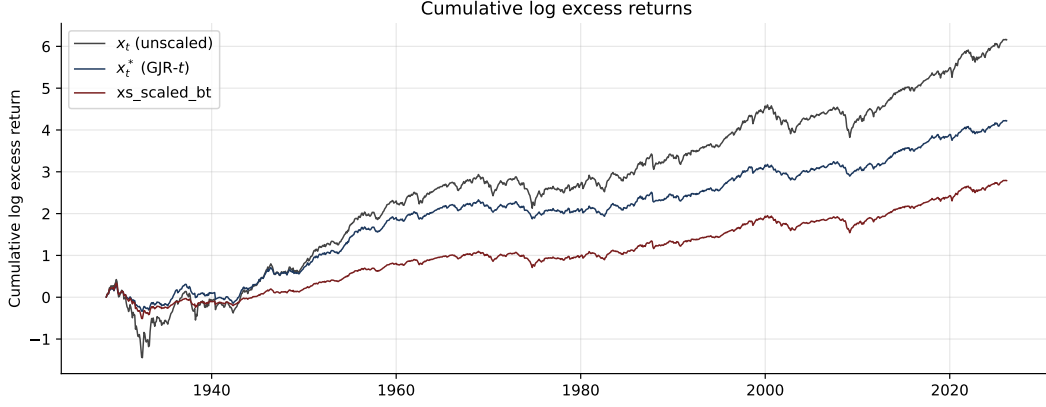


Figure 3: Cumulative log excess returns. The volatility-scaled x_t^* dominates both the unscaled market and the expanding-window benchmark.

5 Discussion

Several caveats apply. (i) The conditional standard deviation $\hat{\sigma}_t$ used to construct x_t^* is extracted from a model estimated on the full sample, so the strategy is not strictly tradeable in real time; this is the same in-sample evaluation used in the assignment appendix. A genuine out-of-sample exercise would re-estimate $\hat{\theta}$ on an expanding window and use one-step-ahead variance forecasts, $\hat{\sigma}_{t|t-1}^2 = \hat{\omega} + (\hat{\alpha} + \hat{\kappa}\mathbb{I}_{\varepsilon_{t-1} < 0})\varepsilon_{t-1}^2 + \hat{\beta}\hat{\sigma}_{t-1}^2$. (ii) Inference is conditional on the GJR- t being correctly specified; if the conditional-mean model is too parsimonious (e.g. a small but real risk-premium component proportional to σ_t) some explanatory power would be miscoded as volatility. (iii) The spanning test is unconditional and does not adjust for transaction costs, financing costs, or leverage constraints required to deliver the targeted 10% annualized volatility in low- $\hat{\sigma}_t$ months. (iv) Jarque–Bera still rejects on $\hat{z}_t$, indicating residual non-normality beyond what $t(7.1)$ captures; this primarily affects extreme-tail probabilities, not the consistency of the QMLE point estimates.

6 Conclusion

U.S. monthly excess returns exhibit strong ARCH effects and statistically significant asymmetric volatility responses. A GJR-GARCH(1,1) with Student- t innovations passes all standard misspecification tests and is the preferred model in our general-to-specific search. Scaling realised returns by its conditional standard deviation raises the annualized Sharpe ratio from 0.34 to 0.43 and produces a positive, HAC-significant alpha against the unscaled market, whereas the expanding-window benchmark in (2.1) of the assignment does not. The evidence supports the volatility-timing thesis of Moreira and Muir (2017) and Harvey et al. (2018), conditional on the in-sample caveats above.